Structural Persistence Theory: Representation Theorem, Second Law, and Formal Verification

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Abstract

We present Structural Persistence Theory (SPT), a framework for measuring how viable structure is irreversibly lost. The theory rests on two conditions: (1) the viable set has positive measure, and (2) repair is never free. A representation theorem, derived from the Cauchy functional equation under normalization, additivity, continuity, and nonnegativity, forces the stage-loss function to be $f(r) = -k \log r$. An impossibility theorem excludes all non-logarithmic alternatives. The persistence kernel $m(V_n) = m(V_0)\, e^{-L_n}$ follows as a telescoping identity. With repair, the net loss $b_t = d_t - r_t$ replaces stage loss, giving $m(V_n) = m(V_0)\, e^{-B_n}$. The structural second law—that cumulative total production Σ_n is monotone nondecreasing—is proved as a necessary and sufficient characterization.

The formalization comprises 394 Lean 4 modules with zero `sorry`/`admit`/`axiom`. Conditional bridges to classical results—Shannon's uniqueness theorem, Jaynes' maximum entropy principle, Landauer's principle, the Crooks fluctuation theorem, and others—are constructed as accounting readouts under domain-specific witnesses. The theory's scope is characterized from both sides: every system satisfying the two conditions admits exactly one loss-measurement form, and every system violating either condition has a formally identified failure mode.

1 Introduction

1.1 The question

Structure can be lost even when resources remain. An organization may retain budget and personnel yet lose the ability to make coherent decisions. Software may retain computational resources yet become unmaintainable as dependency conflicts accumulate. A long-context language model may retain parameters yet lose logical consistency as unresolved contradictions build up.

In each case, what is lost is not the substrate but the set of states that can still sustain the structure. SPT formalizes this observation: structural loss is the shrinkage of the viable set, measured by a log-ratio scale that is uniquely forced by natural axioms.

1.2 Contributions

1. **Representation theorem**: Under normalization, additivity, continuity, and nonnegativity, the stage-loss function must be $f(r) = -k \log r$. No other form is consistent (§3).

2. **Structural second law**: Cumulative total production Σ_n is monotone nondecreasing, proved as necessary and sufficient under two conditions: positive measure and non-free repair (§4).

3. **Formal verification**: 394 Lean 4 modules, zero `sorry`/`admit`/`axiom`, with conditional bridges to 60+ fields (§6).

1.3 What this paper does not claim

- SPT does not claim that all systems decay exponentially. The exponential form is a representation theorem for the viable-set measure, not an empirical law about any specific system.
- SPT does not replace domain-specific dynamics. It provides the accounting coordinates; the dynamics are supplied by each domain.
- The conditional bridges are not unconditional proofs of classical theorems. Each bridge requires domain-specific witnesses (choice of viable set V, measure m, and positivity verification).

2 Setup

2.1 Structural maintenance problem

Fix a system X. A **structural maintenance problem** $\Pi = (X, G)$ consists of:

- A state space X
- A maintenance condition G that defines which states can sustain the structure
- The **viable set** $V_G = \{x \in X : x \text{ satisfies } G\}$—following Aubin's viability theory [1]
- A **measure** m on X, fixed before observation

2.2 Viable-set dynamics

Constraints accumulate over time:

$$V_0 \supseteq V_1 \supseteq \cdots \supseteq V_n.$$

Each step consists of **contraction** (constraints shrink V) followed by **repair** (recovery expands V):

$$V_t^- = K_t(V_t) \qquad \text{[contraction stage]} \tag{1}$$

$$V_{t+1} = R_t(V_t^-) \qquad \text{[repair stage]} \tag{2}$$

2.3 Stage loss and repair

The **stage loss** at step t:

$$d_t = -\log \frac{m(V_t^-)}{m(V_t)}.$$

The **repair amount** at step t:

$$r_t = \log \frac{m(V_{t+1})}{m(V_t^-)}.$$

The **net loss**:

$$b_t = d_t - r_t.$$

2.4 Cumulative quantities and total production

Cumulative stage loss: $L_n = \sum_{t<n} d_t$. Cumulative net loss: $B_n = \sum_{t<n} b_t$. Under a **repair budget** (repair gain $\leq$ repair cost at each step), total production is:

$$\Sigma_n = B_n + C_n = L_n + \text{slack}_n,$$

where C_n is cumulative repair cost and $\text{slack}_n \geq 0$.

3 The Representation Theorem

3.1 Axioms for stage loss

A **stage-loss functional** $f : (0, 1] \to \mathbb{R}$ satisfying:

B2 (Normalization): $f(1) = 0$

B3 (Additivity): $f(r_1 r_2) = f(r_1) + f(r_2)$

B4 (Continuity): f is continuous

- (Nonnegativity): $f(r) \geq 0$ for $r \in (0, 1]$

Theorem 1 (Representation). *Any stage-loss functional satisfying B2+B3+B4+nonnegativity must have*

$$f(r) = -k \log r, \quad k \geq 0.$$

Proof sketch. B3 is the Cauchy functional equation $f(xy) = f(x) + f(y)$. Under B4, the unique continuous solution is $f(r) = c \cdot \log r$. Nonnegativity on $(0, 1]$ forces $c \leq 0$, giving $f(r) = -k \log r$ with $k \geq 0$. Lean: `RepresentationTheorem.loss_must_be_log`. □

Corollary 2 (Persistence kernel). *For any positive mass sequence:*

$$m(V_n) = m(V_0) \cdot e^{-L_n}.$$

At $k = 1$, this is $S = Me^{-L}$. Lean: `TelescopingExp.measure_eq_initial_mul_exp_neg_cumulative_loss`.

Theorem 3 (Impossibility). *No non-logarithmic function satisfies B2+B3+B4+nonnegativity. Linear $a(1 - r)$, quadratic $a(1 - r)^2$, and power ar^α ($\alpha \neq 0$) all violate B3.*
Lean: `ImpossibilityTheorem.impossibility_of_non_log`.

3.2 Relation to Shannon

The representation theorem has the same mathematical skeleton as Shannon's uniqueness theorem for entropy [3]. Both derive from the Cauchy equation; both force a logarithmic form. The difference is in the domain: Shannon measures message uncertainty; SPT measures viable-set shrinkage. SPT does not claim to be Shannon's theorem (see `NonIdentityTheorem`).

4 The Structural Second Law

Theorem 4 (Structural Second Law). *Under (1) positive trajectory ($m(V_t) > 0$ for all t) and (2) repair budget (*gain $\leq$ cost *at each step), cumulative total production Σ_n is monotone nondecreasing.*
Lean: `StructuralSecondLaw.deterministic_second_law`.

Theorem 5 (Converse). *Σ_n monotone $\iff$ stepTotalProduction ≥ 0 at every step.*
Lean: `ConverseSecondLaw.second_law_iff_nonneg_steps`.

The second law holds at three layers:

Layer	Statement	Lean module
Deterministic	Σ_n pointwise monotone	`deterministic_second_law`
Stochastic	$\mathbb{E}[\Sigma_n]$ monotone	`stochastic_second_law`
Coarse-graining	Monotonicity transfers	`coarse_second_law`

Theorem 6 (Free Repair Impossibility). *If* gain > cost *at any step, Σ can decrease. The repair budget is necessary.*

Lean: `FreeRepairImpossibility`.

Theorem 7 (Minimal Axioms). *The two conditions are minimal and jointly sufficient.*

Lean: `MinimalAxiomTheorem`.

4.1 Resource dynamics (M-side)

The resource level M evolves dynamically:

$$M_{n+1} = M_n + \text{income}_n - \text{cost}_n.$$

Theorem 8 (Resource depletion). *If cumulative cost exceeds M_0 plus cumulative income, $M_n < 0$. Lean:* `ResourceDynamics.resource_depleted_when_cost_exceeds`.

Theorem 9 (M-side second law). *Without external income, M is nonincreasing. Lean:* `ResourceDynamics.clc`

Theorem 10 (Dual collapse). *$S_n = M_n \cdot (m_n/m_0)$ collapses when* either *$M_n \leq 0$ (resource exhaustion)* or *$m_n = 0$ (structural death). These are independent failure modes. Lean:* `ResourceDynamics.M_coll`

This completes the $S = Me^{-L}$ picture: both factors are now dynamically tracked.

5 Complete Scope Closure

Condition violated	Failure mode	Lean proof
$m = 0$	Log-ratio undefined	`ScopeBoundaryTheorem`
$m < 0$	Physically meaningless	`CompleteScopeClosure`
gain > cost	Second law violated	`FreeRepairImpossibility`
Constant mass	$L = 0$ (trivial)	`ScopeBoundaryTheorem`

No fifth category exists (`CompleteScopeClosure.classification_exhaustive`).

6 Lean 4 Formalization

The formalization comprises 394 modules (with `sorry` = 0, `admit` = 0, `axiom` = 0. Lean 4 v4.26.0 + Mathlib v4.26.0.

Repository: `https://github.com/karesansui-u/persistence-lean`

Conditional bridges connect the SPT kernel to 60+ fields, including thermodynamics (zeroth through third law readouts), quantum mechanics, statistical mechanics, information theory, probability, biology, economics, engineering, neuroscience, and cosmology. Each bridge is conditional on domain-specific witnesses.

7 Three Observability Layers

Layer	Observability	Role
Specification-fixed	V, m, boundary fixed by specification	Theorems, finite-time bounds
Conditional embedding	Existing theory mapped to SPT variables	Bridges to Foster-Lyapunov, etc.
Structural estimation	Proxy indicators + pre-registered tests	Prediction, diagnosis

These are three observability levels of the same kernel $S = Me^{-L}$.

8 Related Work

SPT is structurally distinct from thermodynamics (allows $B_n < 0$), information theory (different domain), survival analysis (S can exceed 1), and viability theory (adds accounting layer). See `NonIdentityTheorem`.

9 Limitations

SPT provides accounting, not dynamics. The choice of G requires domain knowledge. Core results are finite-horizon. Bridges are conditional.

10 Conclusion

Structural Persistence Theory provides a formally verified framework for measuring irreversible structural loss. Two conditions—positive measure and non-free repair—are necessary and sufficient. The representation theorem forces the unique form $f(r) = -k \log r$. The complete scope closure characterizes applicability from both sides. The Lean 4 formalization with 394 modules and zero `sorry/admit` provides machine-checked confidence.

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